External Enrichment Agent — Design & Verified Research

Status: Phases 1–5 built & tested · **Date:** 2026-08-18
Branch: `feature/physician-facility-data-enrichment`
Builds on: the existing outlook-calendar-poc (OAuth/Graph, Supabase `bis_*` master data, entity-matcher, analytics, `physicianBriefHtml`, email-ingest).

When the rep schedules a meeting with a physician or facility whose email is **not** in `bis_physicians`, the app today shows "*Nobody on this meeting matched the BIS directory*" and stops. This document specifies an agent that fills that gap: it identifies the person from the email alone, pulls their profile from authoritative public sources, **re-matches them back into the BIS tables wherever possible**, and renders a brief in which **every field carries its source** — with explicit badges for what came from BIS, what came from a government registry, what came from the open web, and what is *extra* intelligence that BIS does not model at all.

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1. Guiding principles

1. **BIS data wins, always.** `bis_*` is the master. External data never overwrites it — a disagreement becomes a visible *conflict note*, not a silent replacement. The cascade is designed to extract the maximum from the existing tables *before* and *after* going outside.
2. **Every external claim is attributable.** No field is rendered without a source label, and web-sourced fields carry a clickable proof URL. If we cannot cite it, we do not show it.
3. **Provenance is a first-class type**, not a footnote. Fields are `{value, source, sourceUrl, tier, confidence, retrievedAt}` end to end — store, API, and renderer.
4. **Free registries first, AI last.** NPPES, CMS, Open Payments, PubMed and ClinicalTrials.gov are free and authoritative. The paid AI call is used only for the one job registries cannot do: turning an opaque email address into a name.
5. **Reuse the existing stack.** Same Express routes, same Supabase client, same `app_*` table convention, same brief layout. `bis_*` stays read-only.
6. **Degrade, never crash.** Every source is independently optional — matching the existing store convention (null/empty when a backend is absent).

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2. What the app does today (baseline)

Concern	Today
Physician search	`src/physicians.js → search()` → Supabase RPC `bis_search_physicians`; orphan-facility fallback via `getNearbyForFacility()`
Free-text match	`physicians.matchInText()` (email 100 / name+facility 95 / name 70 / facility 40) and `src/entity-matcher.js → analyze()`
Facility search	**None.** Facilities are loaded once by `bis_directory()` and nested inside each physician; `email-intel.js → facilityInDb()` is an `ilike` existence check
Physician display	One renderer, `graph.physicianBriefHtml()` (`src/graph.js:619`), shared by the briefing email, `/api/physicians/:npi/brief`, `/embed/lead-brief` and `/embed/meeting-brief`
Facility display	4 rows inside `physicianDetailsTable()` (`graph.js:213`); health system + territory derived in `src/territory.js`
ID passing	**NPI is the primary key everywhere** — routes, `app_activities.physician_npi`, notes, `app_email_intel`, `block.dataset.npi`. `facility_id` (e.g. `HSOP105211`) is secondary. Email is a lookup key only
Supabase	RPCs `bis_directory`, `bis_physician_analytics`, `bis_search_physicians`; PostgREST for `app_*`. Anon key, `SUPABASE_ENV` dev/prod switch
Filters	Free-text only — no structured filters (specialty / state / volume)
Frontend	Vanilla JS + `<template>` (`buildDetail` → `buildPhysicianBlock` / `buildNoMatch`)
Auth	MSAL delegated OAuth + `requireAuth` on `/api/*`; token-store for background engines; shared-token + CSP for `/embed/*`; **Supabase RLS is anon-all (POC)**
Caching	Directory in memory once at boot; `product-context` module cache; Redis for sessions/notes. **No cache on analytics, search or brief — and no cache layer for external calls exists yet**

The gap, precisely

- `email-ingest.js → physiciansForEvent()` returns `[]` on a miss ⇒ no activity link, no instant brief, no meeting-body injection.
- `app.js → buildNoMatch()` falls back to a manual search box.

- `lead-match.js` returns `matchedBy: null`.

All three become entry points for the agent.

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3. Verified research (all executed 2026-08-18)

Test subject: `nshaheen@med.unc.edu` — a real academic gastroenterologist who is **not** in `bis_physicians` (the table contains only "Shaheen Rasheed" and "William Shaheen"). Only the email address was supplied as input.

Tier	Source	Result
T0	`bis_physicians.email`	❌ miss
T1	**Claude + `web_search` server tool**	✅ "Nicholas James Shaheen, MD, MPH · Gastroenterology & Hepatology · Chief, Division of GI & Hepatology · UNC" — **confidence 98**, proof URLs `med.unc.edu/cgibd/…`, `med.unc.edu/ppmh/…`, `fda.gov/media/166105` (the email appears verbatim on the UNC pages). 2 searches, ~20.6k in / 1.6k out tokens
T2	**NPPES NPI Registry**	✅ NPI **1467521757**, "NICHOLAS J SHAHEEN, MD", taxonomy *Internal Medicine, Gastroenterology*, 101 Manning Dr, Chapel Hill NC 27599, 919-966-4996, license state NC
T3	**CMS Facility Affiliation** (`27ea-46a8`)	✅ CCN **340061**
T3	**CMS Hospital General Information** (`xubh-q36u`)	✅ UNC HOSPITALS · Acute Care · Government–State · **5-star** · address + phone
T3	**CMS Open Payments 2024**	✅ **29 payments** — Lucid Diagnostics, Exact Sciences, Phathom, Intercept → direct competitor / industry-relationship intel
T3	**PubMed E-utilities**	✅ 337 publications on Barrett's / endoscopy, most recent 2026
T3	**ClinicalTrials.gov v2**	✅ overall official on an RF vapor-ablation trial
T4	**Re-match into BIS**	⚠️ NPI absent from `bis_physicians` — **but the facility is present**: `HSOP105211` "UNC Hospitals Chapel Hill North Carolina", with 6 GI colleagues (emails included) and facility-level CPT volumes (45385 snare polypectomy, 45390 EMR, 45380 biopsy)

Findings that shape the design

1. **An email miss is not a physician miss.** Once NPPES yields an NPI, BIS must be queried *again* by NPI. Even when the physician is genuinely absent, the **facility is often present** — which unlocks the existing volumes, colleagues, territory, health-system and product-fit logic. This is the concrete answer to "*pull as much as possible from my own tables*".

2. **Naive facility matching is dangerous.** A first-cut `ilike` on the first two long words matched "UNC HOSPITALS" to "*ChristianaCare Hospitals Newark Delaware*". Facility matching must filter on **city + state** and score on normalised token overlap, or wrong data reaches the brief.
3. **Email domain → facility is a clean signal in this dataset.** All six `@unch.unc.edu` physicians map to the single facility `HSOP105211`. A domain index built from existing rows is a free, zero-latency pre-tier.
4. **5,179 of 21,274 physicians have no email at all.** The same agent therefore has a second job: backfilling contact gaps in records that *are* in BIS.
5. **NPPES is occasionally flaky** — an identical query returned 0 results once and 1 result moments later. Retry with backoff is mandatory.
6. `data.cms.gov/data-api/v1/dataset/{uuid}/data` (Medicare physician CPT volumes) hangs and times out over both HTTP/2 and HTTP/1.1. Treat as unreliable; ingest from CSV or omit. Every other source above responded in well under 30 s.
7. **NPPES rejects a one-character wildcard.** `first_name=n*` returns HTTP 200 with `{"Errors":[{"description":"Wildcards require at least two leading characters"}]}`. An initial must therefore never be sent to the API — search on the surname alone and rank the results on the initial client-side.
8. **NPPES surname search is fuzzy.** `last_name=Shaheen` genuinely returns people called Williams and Decker (presumably via other-name and authorized-official fields). Unfiltered, this produced the worst result of the whole build: "Evelyn Decker, Counselor, Monterey CA" offered as the match for `nshaheen@med.unc.edu`. The surname must be re-checked client-side.
9. **Two BIS facilities can tie at a perfect score.** "UNC Hospitals Chapel Hill North Carolina" and "UNC Medical Center Chapel Hill North Carolina" both reduce to the single distinctive token `unc`, so the winner came down to iteration order. A secondary, lower-weighted comparison that keeps the generic words ("hospitals" vs "medical center") separates them.

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4. Architecture — a five-tier, DB-first cascade

```
Meeting attendee email (unknown to BIS)
    │
T0   BIS DB           email → physician?                  FREE, 0 ms   → hit: stop, existing
path
    │ miss
T0.5 Domain index     @unch.unc.edu → HSOP105211          FREE, 0 ms   (built from existing
rows)
    │
T1   Identity resolve  Claude + web_search → name, title,    ~$0.15      (the only paid step)
    │                  institution, state + PROOF URLs
T2   Registry          NPPES: name+state → NPI, taxonomy,     FREE
    │                  practice address, phone, license
T3   RE-MATCH TO BIS   by NPI, and by facility(city+state+    FREE         ← maximise own data
    │                  token overlap)
T4   Public enrichment CMS affiliation → CCN → hospital;      FREE
    │                  Open Payments; PubMed; ClinicalTrials
    │
T5   Persist + render  app_external_profiles → provenance-tagged brief
```

Ordering rationale. T2–T4 are free and fast, so they run in parallel and render immediately; T1 is slow (15–25 s) and paid, so its result streams into the card afterwards rather than blocking the first paint.

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5. The provenance model

Every field is an object, not a scalar:

```
{
  value: "Gastroenterology",
  source: "NPPES NPI Registry",
  sourceUrl: "https://npiregistry.cms.hhs.gov/provider-view/1467521757",
  tier: "verified",            // db | verified | web | inferred
  confidence: 100,
  retrievedAt: "2026-08-18T09:14:22.000Z"
}
```

Four tiers, four badges

Badge	Tier	Meaning	Sources
🟢 **BIS**	`db`	From your own Supabase master	`bis_physicians` , `bis_facilities` , `bis_procedure_volumes`
🔵 **Verified**	`verified`	Official government registry	NPPES, CMS Care Compare, CMS Open Payments
🟡 **Web**	`web`	AI-extracted from the open web — **always with a link**	Hospital bio pages, faculty directories, Doximity, news
⚪ **Inferred**	`inferred`	Derived by us	email domain → org, state → territory, name → health system

Two explicit flags

- `+ EXTRA` — the field has no counterpart in the `bis_*` schema (publications, hospital star rating, industry payments, academic title, clinical trials). These are grouped under a separate **"Extra Intelligence (not in BIS)"** heading so the rep can tell instantly what is new versus what is enriched.
- **Origin banner** at the top of the card:

> ⚠️ **This physician is not in your BIS database.** Profile assembled from external sources. and, when a facility match succeeds: ✅ **Facility found in BIS:** UNC Hospitals (`HSOP105211`) — volumes and colleagues below come from BIS.

Each external field renders a small `🔗` linking to its exact source page. In the email version this becomes a plain `<a>` so it survives Outlook rendering.

Precedence and conflicts

`db > verified > web > inferred`. When two sources supply the same field, the higher tier wins and the loser is retained as a conflict note:

> *NPPES reports specialty "Internal Medicine, Gastroenterology"; BIS holds "Gastroenterology".*

Nothing is silently overwritten, in either direction.

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6. Modules

```
src/enrichment/
  index.js              orchestrator — runs the cascade, returns a ProvenanceProfile
  provenance.js         field() helper, tier precedence, merge + conflict capture
  rematch.js            NPI re-lookup + facility match (city+state+token overlap)
  cache.js              app_external_profiles read/write + in-memory LRU + TTL
  sources/
    nppes.js            NPI registry, individuals (NPI-1) and organisations (NPI-2)
    cms-provider.js     facility affiliation (27ea-46a8) + hospital info (xubh-q36u)
    open-payments.js    industry payments by NPI (competitor intelligence)
    literature.js       PubMed E-utilities + ClinicalTrials.gov v2
    web-identity.js     Claude web_search → citations → forced-tool-use JSON
src/routes/api.routes.js  + GET /api/enrich, POST /api/enrich/:npi/promote
src/graph.js              + externalBriefHtml() — provenance-aware sibling of physicianBriefHtml
supabase/enrichment-setup.sql
```

Source endpoints (verified working)

Source	Endpoint	Key
NPPES	`https://npiregistry.cms.hhs.gov/api/?version=2.1&last_name=&first_name=&state=&limit=`	none
NPPES orgs	same + `enumeration_type=NPI-2&organization_name=`	none
CMS affiliation	`https://data.cms.gov/provider-data/api/1/datastore/query/27ea-46a8/0?conditions[0][property]=npi&conditions[0][value]=<NPI>&conditions[0][operator]==`	none
CMS hospital	`…/datastore/query/xubh-q36u/0?conditions[0][property]=facility_id&conditions[0][value]=<CCN>&conditions[0][operator]==`	none
CMS hospital by name	same with `&conditions[0][property]=facility_name&conditions[0][value]=UNC%&conditions[0][operator]=like`	none
Open Payments 2024	`https://openpaymentsdata.cms.gov/api/1/datastore/query/e6b17c6a-2534-4207-a4a1-6746a14911ff/0?conditions[0][property]=covered_recipient_npi&…`	none
PubMed	`https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=<Last>+<Initials>[Author]` then `esummary.fcgi`	none
ClinicalTrials	`https://clinicaltrials.gov/api/v2/studies?query.term=&fields=NCTId,BriefTitle,OverallOfficialName`	none
Identity	Anthropic Messages API, `web_search` server tool, then a second forced-tool-use call for strict JSON	`ANTHROPIC_API_KEY` (already set)

> **PubMed gotcha:** the author term must use initials — `Shaheen NJ[Author]`. A bare `Shaheen N` returns unrelated papers across all of medicine.

The two-call AI pattern

The repo already uses forced tool-use for strict JSON (`ai-extractor.js`, `intel-extractor.js`). Forcing `tool_choice` would prevent the model from searching, so identity resolution is split:

1. **Call 1** — `web_search` enabled, `tool_choice: auto`. Returns prose plus per-sentence citations (`{url, title}`).
2. **Call 2** — forced tool use over call 1's text + citation list, producing `{is_physician, full_name, first_name, last_name, credentials, specialty, title, institution, city, state, confidence, evidence_urls}`.

Reuse `intel-extractor.js`'s `sanitize()` pass — the same delimiter-leak gotcha documented in `CLAUDE.md` applies here.

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7. Data model

```
create table if not exists public.app_external_profiles (
  id                   uuid primary key default gen_random_uuid(),
  lookup_email         text,          -- what we searched by
  resolved_npi         text,          -- from NPPES (nullable)
  in_bis               boolean not null default false,
  matched_facility_id text,          -- bis_facilities hit, if any
  profile              jsonb not null,  -- full provenance-tagged profile
  sources              jsonb not null,  -- [{source, url, fields[], retrievedAt}]
  confidence           int,
  created_at           timestamptz default now(),
  refreshed_at         timestamptz default now()
);
create unique index if not exists app_external_profiles_email_idx
  on public.app_external_profiles (lower(lookup_email));
create index if not exists app_external_profiles_npi_idx
  on public.app_external_profiles (resolved_npi);
```

14-day TTL on `refreshed_at`, then background refresh. Follows the existing `app_*` convention; DDL is run by hand in the Supabase SQL editor, per `CLAUDE.md`. Dev first, prod before go-live.

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8. Integration points

Location	Today	After
`app.js → buildNoMatch()`	"Nobody matched — pick who the meeting is with" + search box	Auto-enrich first; render the external card with badges. Search box becomes the fallback
`email-ingest.js → physiciansForEvent()`	miss ⇒ `[]` , no brief	miss ⇒ enqueue enrichment; send the instant brief with an `[EXTERNAL]` marker once resolved
`graph.physicianBriefHtml()`	unchanged	new sibling `externalBriefHtml()` — same layout plus badges and the Extra Intelligence block. The email == in-app invariant is preserved
`lead-match.js`	`matchedBy: null`	`matchedBy: 'external'` — Dynamics leads enrich too
`email-intel.js` "New to DB?"	flags only	flags **and** offers one-click enrichment
new `POST /api/enrich/:npi/promote`	—	rep-verified profile is written to `app_contacts` ; `bis_*` stays read-only

Facility-only enrichment

When the attendee is a generic mailbox (`info@somegi.com`) or no physician resolves:

1. Domain → organisation name (web search).
2. **NPPES organisation search** (`enumeration_type=NPI-2`) — verified working.
3. **CMS Hospital General Information name search** (`operator=like`) — verified working.
4. Match against `bis_facilities` on city + state + token overlap.
5. On a hit, surface that facility's BIS physicians and facility-level CPT volumes.

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9. Cost, latency, caching

- **Web search:** $10 per 1,000 searches. On `claude-opus-4-8` ($5 / $25 per MTok) the measured call was ~20.6k in / 1.6k out plus 2 searches ⇒ **≈ $0.15–0.20 per new physician.** On Sonnet-tier, ≈ $0.07.
- **Every other source is free** and needs no API key.
- **Latency:** free tiers ≈ 5 s combined; the web tier 15–25 s. Render free-tier results first, stream the identity block in.
- **Caching** (new, since none exists today): `app_external_profiles` keyed by lower(email) with a 14-day TTL, plus a small in-memory LRU per process. Repeat lookups cost nothing.
- **Guardrails:** per-day enrichment budget, internal-domain skip-list, and an early exit when the model reports `is_physician: false` .

10. Delivery phases

Phase	Scope	Cost
P1 ✅	`sources/nppes.js`, `sources/cms-provider.js`, `rematch.js`, `provenance.js`, `states.js`, `http.js`, `GET /api/enrich` — **built and tested 2026-08-18**, see §13	**$0 — no AI**
P2 ✅	`sources/web-identity.js` (web_search + citations + forced-tool-use JSON), `context.js` (organizer rule + title/description hints), confidence gating — **built and tested 2026-08-18**, see §14	paid tier begins
P3 ✅	`externalBriefHtml()` + badges + Extra Intelligence block; `buildNoMatch()` auto-enrich — **built and tested 2026-08-18**, see §15	—
P4 ✅	`app_external_profiles` cache + TTL; Open Payments, PubMed, ClinicalTrials extras — **built and tested 2026-08-18**, see §16	—
P5 ✅	`email-ingest` hook (unknown attendee ⇒ auto brief), `promote` flow, backfill script — **built and tested 2026-08-18**, see §17	—

P1 and P2 need no new credentials — `ANTHROPIC_API_KEY` is already configured and `web_search` is enabled on it.

11. Open decisions

1. **Confidence threshold.** 98 % is unambiguous; 60 % (common surname, small private practice) is not. Proposed: **auto-show at ≥ 70, otherwise present "possible matches — confirm" chips** — mirroring the existing entity-matcher suggestion UX.
2. **Write policy.** Proposed: enrichment **never** writes to `bis_*`. It lives in `app_external_profiles`; only rep-verified data is promoted to `app_contacts`.
3. **Auto vs on-demand.** Proposed: **automatic** for calendar meetings (the rep should not have to ask), **on-demand** for Leads and the Email Sheet. Cost is already bounded by the P2 gating (§14) — the paid tier only fires when there is genuinely something to buy.

4. **Non-physician handling.** ✅ **Decided 2026-08-18.** No internal-domain skip-list — an exclusion by ROLE instead: **only attendees are matched, and the organizer (the rep who scheduled the meeting) is never matched, ever.** Meeting-room resources are excluded the same way. Non-clinicians additionally exit at `is_physician: false`, which is evidence-based rather than list-based, so nothing has to be configured or kept up to date.

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12. Risks

Risk	Mitigation
Wrong person matched (common name)	Confidence gate; require NPPES corroboration of state/specialty before promoting to `verified`; show proof URLs so the rep can check in one click
Wrong facility matched	City + state filter plus token-overlap scoring — the naive `ilike` failure is documented in §3
NPPES transient empty results	Retry with exponential backoff; treat 0-results as retryable, not authoritative
Cost runaway	Cache + TTL, per-day budget, skip-list, early `is_physician` exit
Stale external data	14-day TTL with background refresh; `retrievedAt` rendered on every external field
Source markup / schema drift	Each source module is isolated and independently optional; a failing source omits its section rather than failing the profile
PHI / compliance	All sources are public government or public-web data; no patient data touched. `app_external_profiles` inherits the existing POC RLS caveat and must be tightened with the rest before production

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13. Phase 1 — what shipped

Built and verified 2026-08-18 on branch `feature/physician-facility-data-enrichment`. No AI, no new credentials, no new dependencies; `bis_*` untouched.

```
src/enrichment/
  index.js              orchestrator: T0 → T0.5 → T2 → T3 → T4
  provenance.js         field(), tier precedence, conflicts, dropWhere()
  rematch.js            domain index, email name-hints, facility matching
  states.js             state code ↔ full name (bis_* holds names, CMS/NPPES codes)
  http.js               timeout + backoff + never-throw fetch wrapper
  sources/nppes.js      individuals, organisations, by-NPI
  sources/cms-provider.js  affiliation → CCN → hospital, hospital name search
src/routes/api.routes.js  GET /api/enrich
```

API

`GET /api/enrich?email=&name=&firstName=&lastName=&state=&city=&npi=&facility=` (session-authenticated, like every other `/api/*` route). At least one of `email`, `name`, `lastName`, `npi` is required.

`status`	Meaning
`in_bis`	Already in `bis_physicians` — use the existing brief, no enrichment
`recovered_in_bis`	Email was missing from the master, but the resolved NPI is in it
`external`	Genuinely outside BIS; registry profile returned (confidence ≥ 70)
`ambiguous`	Best match is 40–69 % — show as "possible match, confirm"
`facility_only`	Person unresolved, facility identified
`unresolved`	Nothing confident enough; candidates listed under `alternatives`

The response carries `profile.fields`, `profile.extra` (the `+ EXTRA` group), `profile.conflicts`, `profile.notes`, `profile.sources`, plus `matchedFacility`, `colleagues`, `alternatives`, `confidence`, `tiers` and `elapsedMs`.

Measured results

Case	Input	Result
A	`lisa.gangarosa@unch.unc.edu` (in BIS)	`in_bis`, 100 %, **0 ms**
B	`nshaheen@med.unc.edu`, no hints	`unresolved` — 20 Shaheens nationwide, none confident. 4 alternatives listed. 5.4 s
C	same + `state=NC`	`external`, **NPI 1467521757**, 71 %, facility → **HSOP105211** with the 6 GI colleagues. 1.9 s
D	`name=Nicholas Shaheen, MD&state=NC`	`external`, 100 %, same facility. 1.0 s
E	`jgreenwood@unch.unc.edu` (unknown person, known domain)	`ambiguous` 46 % + 3 alternatives; facility still resolved from the domain index. 7.1 s
F	`info@unch.unc.edu` (generic mailbox)	`facility_only`, 90 %, **3 ms** — recognised as an organisational address
G	`npi=1467521757`	`external`, 100 %, 1.0 s
H	`zzqqxx@nowhere-clinic-xyz.com`	`unresolved`, no false claim. 4.7 s

Case C is the design's central claim, demonstrated end to end: a physician who is **not** in `bis_physicians` is identified, and the brief is still populated from BIS — the right facility, its territory and health system, and six real colleagues — because the facility *is* in the master.

Candidate lookups run in parallel, which cut the worst case (H) from 17 s to 4.7 s.

Known limits (addressed by P2)

- Without a state hint, a common surname cannot be resolved from the email alone — case B. The web identity tier reads the name and institution off the person's own page, which supplies exactly the missing hint.
- Only institutional address conventions are parsed (`first.last@`, `nshaheen@`). An opaque local part yields nothing until P2.
- Nothing is cached yet: every call re-queries the registries (P4).

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14. Phase 2 — what shipped

Built and verified 2026-08-18. Adds the paid identity tier and the matching rules the rep set.

```
src/enrichment/sources/web-identity.js   T1 — email → name, via Claude + web_search
src/enrichment/context.js                who to enrich + title/description hints
src/email-ingest.js, src/reminders.js    organizer rule enforced
src/routes/api.routes.js                 organizer rule + ?context= & ?useWeb=
```

The matching rules

Only attendees are matched. The organizer is never matched — ever. The organizer is the rep who scheduled the meeting; briefing them on themselves, or spending an AI lookup on them, is always wrong. This is enforced in one place (`context.attendeesToEnrich`) and applied at every site that resolves a physician: the calendar API, the ingest engine's auto-brief and meeting-body injection, and the reminder engine. Meeting rooms (`type: 'resource'`) and the signed-in user are excluded by the same rule. **This replaces the internal-domain skip-list** — an exclusion by role needs no configuration and cannot go stale.

The title and description are context, not identity. They supply the facility, city and other details the organizer typed in — resolved through the existing entity-matcher, so a confident hit returns a real `bis_facilities` row with its city and state. Person entities found there are ignored for identification, and any that are really the organizer's own name are stripped before the analysis is used anywhere.

The web identity tier

Two calls, because they cannot be one: `web_search` needs `tool_choice: auto` (forcing a tool would prevent searching), then a second forced-tool-use call turns the findings into strict JSON — the repo's established pattern, including the `sanitize()` delimiter-leak pass from `CLAUDE.md`.

It uses the basic `web_search_20250305`, not the newer `web_search_20260209`. Measured head-to-head on the same prompt and model:

Tool version	Time	Citations
`web_search_20250305`	**14 s**	**7 (4 unique proof URLs)**
`web_search_20260209`	200 s+, once a 396 s timeout	**0**

The newer variant filters results through code execution, which appears to cost the per-sentence citations. Citations are not a nice-to-have here — a web-sourced field with no link to prove it is exactly what this design refuses to render.

Cost gating

The paid call fires only when there is something to buy. It is skipped when the caller already supplied a name, when the mailbox is organisational (`info@` , `scheduling@` …), when `ANTHROPIC_API_KEY` is unset, or when the caller passes `useWeb=never`. `useWeb=always` forces it. A confident `is_physician: false` ends the enrichment immediately — before any registry call.

Measured results

Case	Result	Paid call
`nshaheen@med.unc.edu` , **email only, no hints**	`external` , **98 %**, NPI 1467521757, facility **HSOP105211** + 6 GI colleagues, 4 proof URLs. 41 s	yes — 2 searches, 24.4k in / 2.1k out
same, `useWeb=never`	`unresolved` (P1 behaviour, unchanged), 5.1 s	no
`info@unch.unc.edu`	`facility_only` , 90 %, 3 ms	no
`name=Nicholas Shaheen&state=NC`	`external` , 100 %, 1.0 s	no
`abdul@primathon.in` (real non-clinician)	`not_physician` — "Technology Advisor / Director", no brief produced. 33 s	yes, then stopped
Meeting with organizer + physician + room + colleague	enriches **only** the two non-organizer people; room and organizer dropped	—

The first row is the case Phase 1 could not solve: with no state hint, twenty Shaheens nationwide made the registry ambiguous. The web tier reads the name and institution off the person's own faculty page — including a page that prints the email address verbatim — and that supplies the missing hint. Evidence recorded with it: "*The email nshaheen@med.unc.edu appears directly next to Nicholas J. Shaheen…*"

Known limits

- 41 s end-to-end when the paid tier runs. The free tiers resolve in 1–5 s, so P3's UI should render those first and stream the identity block in.
- For a non-physician the *identity* may be imprecise (it only has to be confident that the person is not a clinician) — nothing is rendered from it.
- Still no caching: every call re-queries (P4).

---

15. Phase 3 — what shipped

Built and verified 2026-08-18. The enrichment is now visible in the app, and the title-based person fallback is gone.

```
src/graph.js              + externalBriefHtml()  (sibling of physicianBriefHtml)
src/routes/api.routes.js  /api/enrich now returns rendered `html`
public/app.js             buildEnrichment() — auto-lookup inside a meeting
public/styles.css         .enrich* card framing
src/email-ingest.js       title→person fallback REMOVED
src/reminders.js          title→person fallback REMOVED
```

Identity is now attendee-email only

Both engines previously fell back to matching a PERSON out of the meeting title/description when no attendee email matched. That is removed. These are the functions that send the automatic brief email and inject the brief into the meeting body, so a wrong guess mails the wrong physician's data to the rep. The reminder engine still honours the physician the rep **explicitly scheduled** with (`app_activities`), which is a recorded choice rather than a guess, and the title/description still supply facility context — just never an identity.

The rendered brief

`externalBriefHtml(result)` is a deliberate sibling of `physicianBriefHtml` , not a branch inside it: this brief carries things the BIS one never does.

- **Origin banner** — ⚠️ not in BIS / ✅ recovered from BIS / 🚫 non-physician, plus a green banner naming the BIS facility when one was matched, so the rep can see at a glance which data is their own.
- **Per-field provenance** — every row shows its badge (🟢 BIS · 🔵 Verified · 🟡 Web · ⚪ Inferred), its source name, and a 🔗 to the page that proves it.
- **Extra Intelligence** — fields BIS has no column for, under a `+ EXTRA — not held in BIS` tag: title, institution, licence, CMS star rating, ownership, CCN, and the clickable evidence URLs.
- **Source disagreements** — rendered, never silently resolved. A live example: *kept **NICHOLAS J SHAHEEN** (NPPES) over "Nicholas James Shaheen" (Web)*; *kept **UNC Hospitals Chapel Hill North Carolina** (BIS) over "UNC HOSPITALS" (CMS)*.
- **Colleagues at this facility** — from BIS, the payoff when the person themselves is absent from the master.
- **Sources footer** — one line per source listing exactly which fields it fed.
- Headings adapt: with no person resolved the table is titled *Facility details*, not *Physician details*.

In the app

Opening a meeting whose attendees are not in BIS now runs the lookup automatically, one card per attendee **— and never for the organizer**, filtered on the client as well as the server.

The lookup is two-stage on purpose: the free tiers answer in 1–5 s and run automatically, and the ~40 s paid web tier sits behind a **"🔎 Identify with web search"** button that only appears when the free tiers actually fell short (`unresolved` , `facility_only` , `ambiguous`). The rep sees something immediately, and no meeting costs money just by being opened.

`/api/enrich` returns the rendered `html` alongside the JSON — the same convention `/api/physicians/:npi/brief` and `/api/leads/match` already follow. A physician already in BIS gets the **standard** brief (nothing about them is enriched); a recovered one gets the origin banner followed by their full BIS brief.

16. Phase 4 — what shipped

Built and verified 2026-08-18: the depth sources, and a cache so the expensive work happens once.

```
src/enrichment/sources/open-payments.js   industry payments by NPI
src/enrichment/sources/literature.js      PubMed + ClinicalTrials.gov
src/enrichment/cache.js                   app_external_profiles + in-process LRU
supabase/enrichment-setup.sql             the table (run by hand, per CLAUDE.md)
```

Industry payments — the competitor intelligence

This is the most commercially useful thing the agent finds outside BIS, and BIS has no column for any of it. Live result for the test physician:

> **$46,990 across 40 payments from 5 companies (2024–2025)**, most recent 25 Oct 2025 — Phathom ($24,217 over 25 payments), Intercept ($11,250), **Lucid Diagnostics** ($5,970), **PENTAX of America** ($3,000), **Medtronic** ($1,565). Named products include **BARRX** and **NEXPOWDER**.

A rep walking into that meeting now knows which device companies are already paying this physician, how much, how recently, and for what — before saying a word. Two yearly datasets are queried in parallel (2025 and 2024); both use the same `covered_recipient_npi` column.

Publications and trials

PubMed's author index is `Surname II` — **initials, not a first name**. "Shaheen Nicholas" matches nothing and a bare "Shaheen" matches every Shaheen in medicine, so `authorTerm()` builds the initialled form: **458 publications** for `Shaheen NJ[Author]`, most recent 2026 in *Aliment Pharmacol Ther*. ClinicalTrials.gov matches a term anywhere in a study, so hits are kept only when the person is actually named as an overall official — **4 trials**, and a nonsense name correctly returns nothing.

Together these separate a high-volume community endoscopist from a KOL who publishes and runs trials on the exact procedure the product serves.

The cache

`app_external_profiles`, keyed by lowercased email, holding the **whole** enrich() result so a hit rebuilds the brief byte-for-byte — provenance included — plus a small per-process layer in front of it. Default TTL 14 days (`ENRICHMENT_CACHE_DAYS`), `?refresh=1` forces a fresh lookup.

Three rules keep it from doing harm:

1. **Never serve a shallower answer than the caller asked for.** A free-tier result must not satisfy a request that wants the paid tier, or "🔎 Identify with web search" would return the same *unresolved* it was pressed to fix.
2. **Never cache a BIS hit.** `in_bis` is a free in-memory lookup that must reflect the master as it is now.

3. **Never cache a failure.** This one was written *because* of a real outage during testing (below): an `unresolved` is as often a transient source problem as a real dead end, and remembering it would have hidden a real physician for a fortnight. `ambiguous` is kept only when the paid tier already ran, since retrying would spend that money to reach the same weak answer.

The table is created by hand (`supabase/enrichment-setup.sql`), so until it is run the cache logs one warning and falls back to in-process only. PostgREST reports a missing table as `PGRST205` / "Could not find the table … in the schema cache", **not** Postgres's `42P01` — matching only on the latter made it log a failure on every single lookup instead of disabling itself once.

A real outage, mid-test

Part way through verification, `npiregistry.cms.hhs.gov` stopped resolving — `SERVFAIL` from the local router's DNS (192.168.31.1), while 8.8.8.8 resolved it fine and every other source stayed at HTTP 200. So NPPES itself was up; the local resolver was not.

The agent behaved exactly as designed: three retries with backoff, one warning per source, the cascade continued, and the request returned `unresolved` instead of throwing. It also produced the failure that motivated cache rule 3.

Worth knowing operationally: **the running app depends on this hostname**, so a DNS problem on the deployment host degrades enrichment to the web and CMS tiers only — without breaking anything.

---

17. Phase 5 — what shipped

Built and verified 2026-08-18. The agent now runs without being asked, and what it finds can become part of the app's own records.

```
src/email-ingest.js          briefUnknownAttendees() — the hook
src/graph.js                 + sendExternalBriefing()
src/routes/api.routes.js     POST /api/enrich/promote
scripts/enrich-backfill.js   npm run enrich:backfill
src/enrichment/cache.js      addressable by email OR npi
```

The ingest hook — the gap finally closed

A meeting whose attendees were not in `bis_physicians` used to end in silence: no activity link, no brief, no meeting-body injection. The rep walked in with nothing. `syncActivities` now enriches those attendees, and two outcomes matter:

- `recovered_in_bis` — the physician IS in the master; only their email was missing. The meeting is linked to their NPI and briefed through the **normal** path, exactly as if the address had matched. This alone rescues meetings that were previously invisible to the whole platform.

- `external` — genuinely outside BIS. The rep gets the provenance-tagged brief by email (`sendExternalBriefing`), rendered from the same `externalBriefHtml` as the in-app card, so email and app still match.

Guards: never the organizer, once per meeting (dedup key `enrich:<eventId>`), and only for meetings not seen before — so a five-minute poll cannot spend repeatedly on the same event. Results are cached for 14 days on top of that.

Promote — how enrichment becomes the app's own data

`POST /api/enrich/promote` writes confirmed contact details into `app_contacts`, the Contact Intelligence overlay the brief already reads, with provenance recorded in `source` (`enrichment:… (confirmed by <rep>)`).

Deliberately manual. The agent writes to its own cache automatically, but a person decides what is trustworthy enough to keep — and `bis_*` is still never written to.

Backfill

```
npm run enrich:backfill -- --facility HSOP105211 --limit 6
npm run enrich:backfill -- --missing-email --limit 20 --write
npm run enrich:backfill -- --npi 1467521757,1508935800 --write
```

Pre-computes the outside-BIS picture for a set of physicians so the first meeting is instant. Defaults are conservative on purpose — free tiers, 20 physicians, nothing written — because a blanket sweep of all 5,179 email-less physicians through the paid tier would cost roughly **$780**. That should be a deliberate decision, not the result of running a script with no arguments.

A live run over one facility's six gastroenterologists:

```
1/6 1508935800 Lisa M Gangarosa — recovered_in_bis · 12 pubs
2/6 1194760868 Julia W Tang     — recovered_in_bis · $474 from 5 co.
3/6 1336534387 Andrew J Gilman  — recovered_in_bis · 2 pubs · 1 trial
4/6 1922234863 Animesh Jain     — recovered_in_bis · 6 pubs
5/6 1508126079 Craig Reed       — recovered_in_bis · 1 pub
6/6 1588954457 Neil D Shah      — recovered_in_bis · $26,959 from 5 co. · 4 pubs
```

The cache is now addressable by **NPI as well as email** (`lookup_key` is either the address or `npi:<npi>`), because a physician who is in BIS with no email has no address to key on.

A precision bug the backfill exposed

The first run reported **"Animesh Jain — 10,279 publications"**. `Jain A[Author]` is not a person; it is thousands of them. Presenting that to a rep as one physician's output is worse than reporting nothing at all.

PubMed author searches are now narrowed by the physician's own affiliation **and** specialty, taken from the NPPES record already in hand:

	Before	After
Animesh Jain (common name)	10,279	**18**
Nicholas Shaheen (distinctive name)	458	**275**

Affiliations are alternatives to each other but the specialty is an additional requirement — OR-ing them together let every "Jain A" paper about gastroenterology back in from anywhere in the world. Where no narrowing signal exists the count is still shown, labelled "*surname match only, not verified as this person*", and the exact search term travels with the field so the rep can click through and check.

This is the same principle as the confidence bands and the conflict list: the brief may say "we don't know", but it must never say something confident and wrong.